

华中科技大学

Huazhong University of Science and Technology



华中科技大学

网络空间安全学院

School of Cyber Science and Engineering, HUST

实验二 机器人连接和远程控制

华中科技大学 2021 ROS创新实践教学



日期：2021.03



内容：Turtlebot3远程控制

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概述

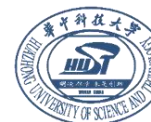
连线 并通电



配置 网络

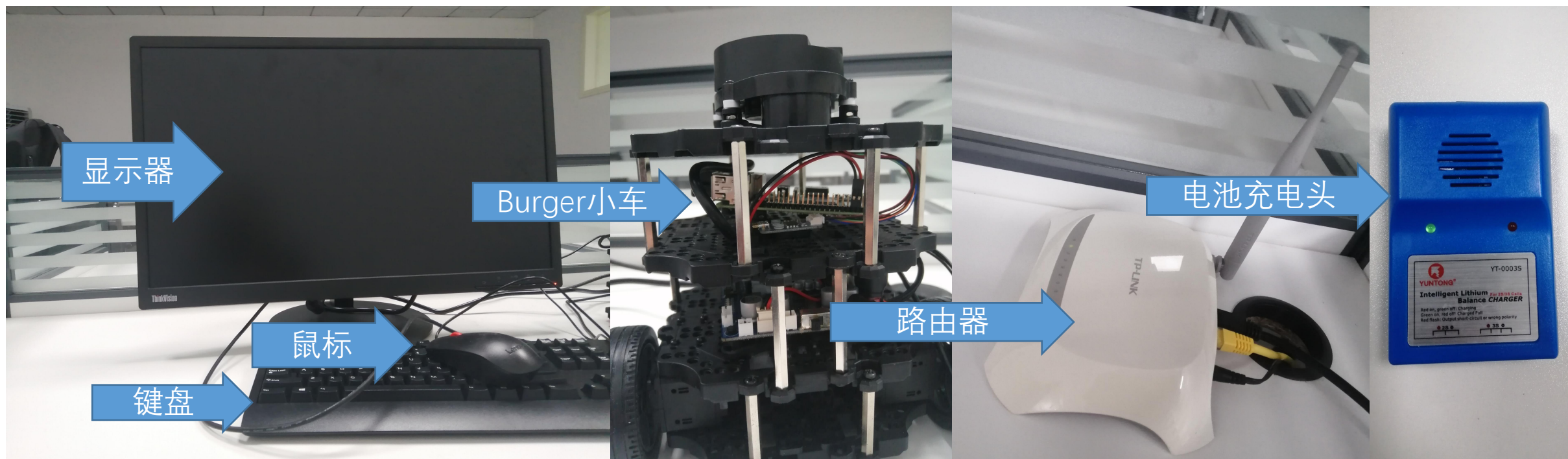


远程 操控真小车

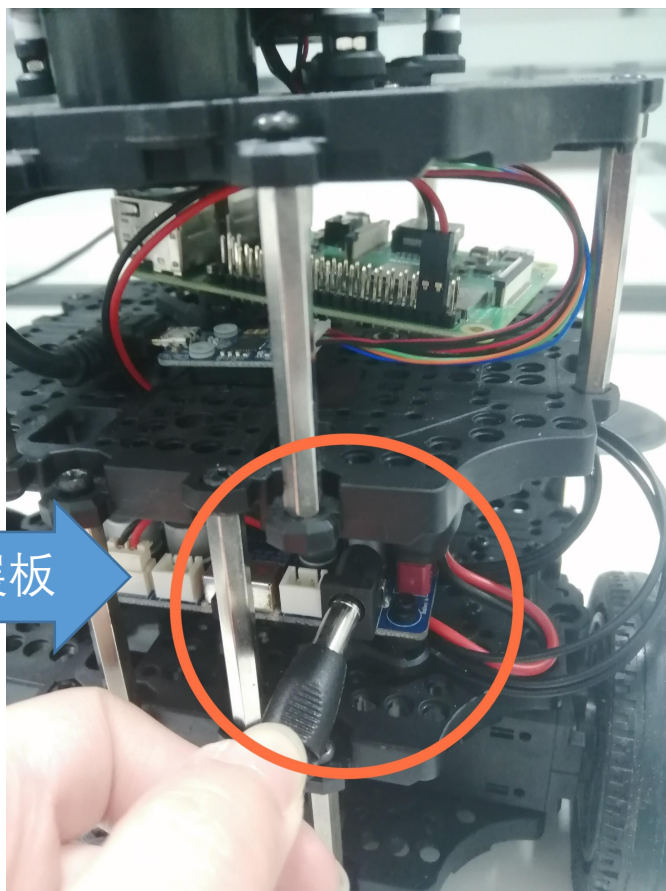


本关涉及的设备：

显示器、Burger小车、路由器、HDMI线、网线、电源适配器、电池充电头等。



供电分**直接供电**和**电池供电**，可以直接用电源适配器供电（如左图），也可以用电源适配器+充电头给电池充电（如右图）。



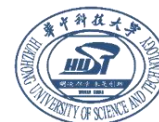
扩展板



电源适配器

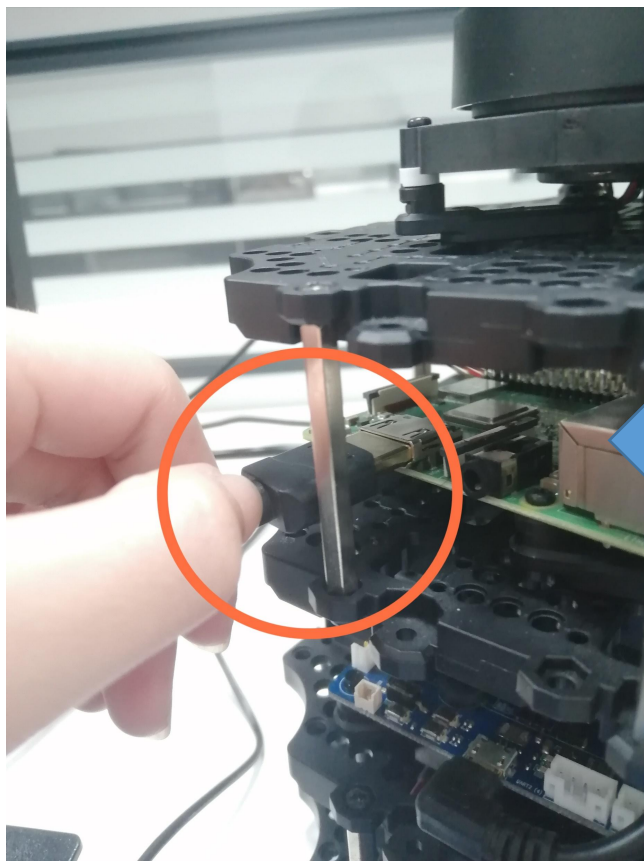
建议本实验
直接供电

扩展板下面一层的电池



第一关——连接HDMI线

HDMI线一头接在Burger的树莓派上面那一层（如左图），另一头接在显示器背面（如右图），连接好显示器的电源并通电。

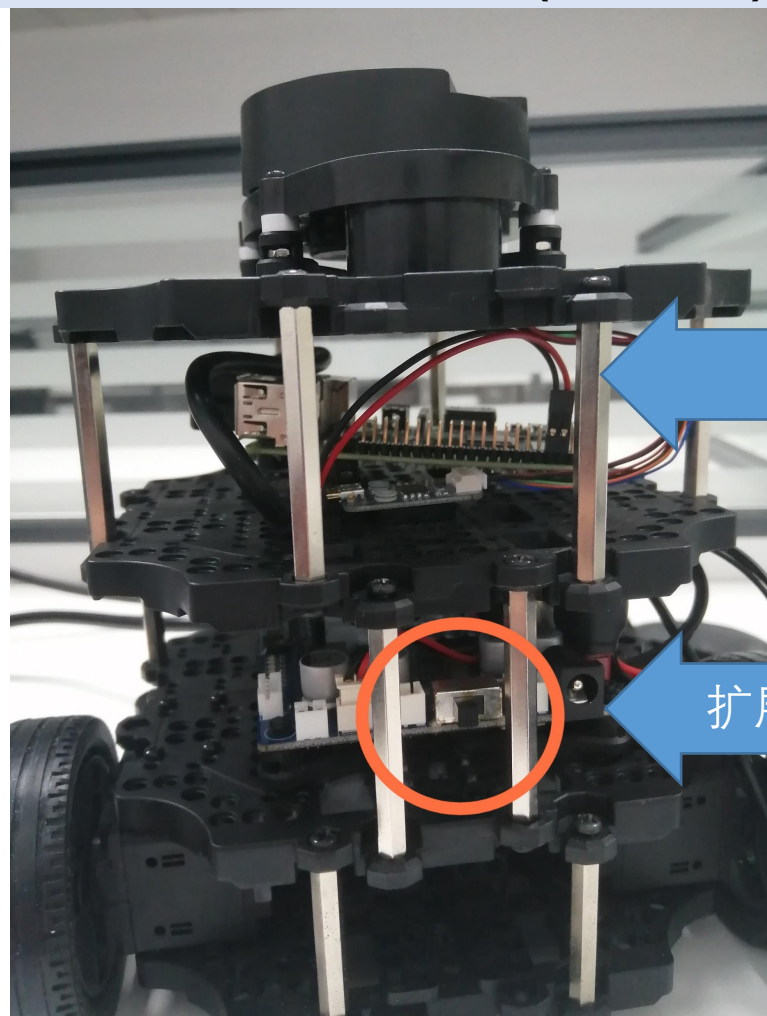


树莓派

HDMI线



打开树莓派上的开关（如左图）。并连接鼠标和键盘（树莓派的上一层的USB接口）。



树莓派

扩展板上的开关

结果

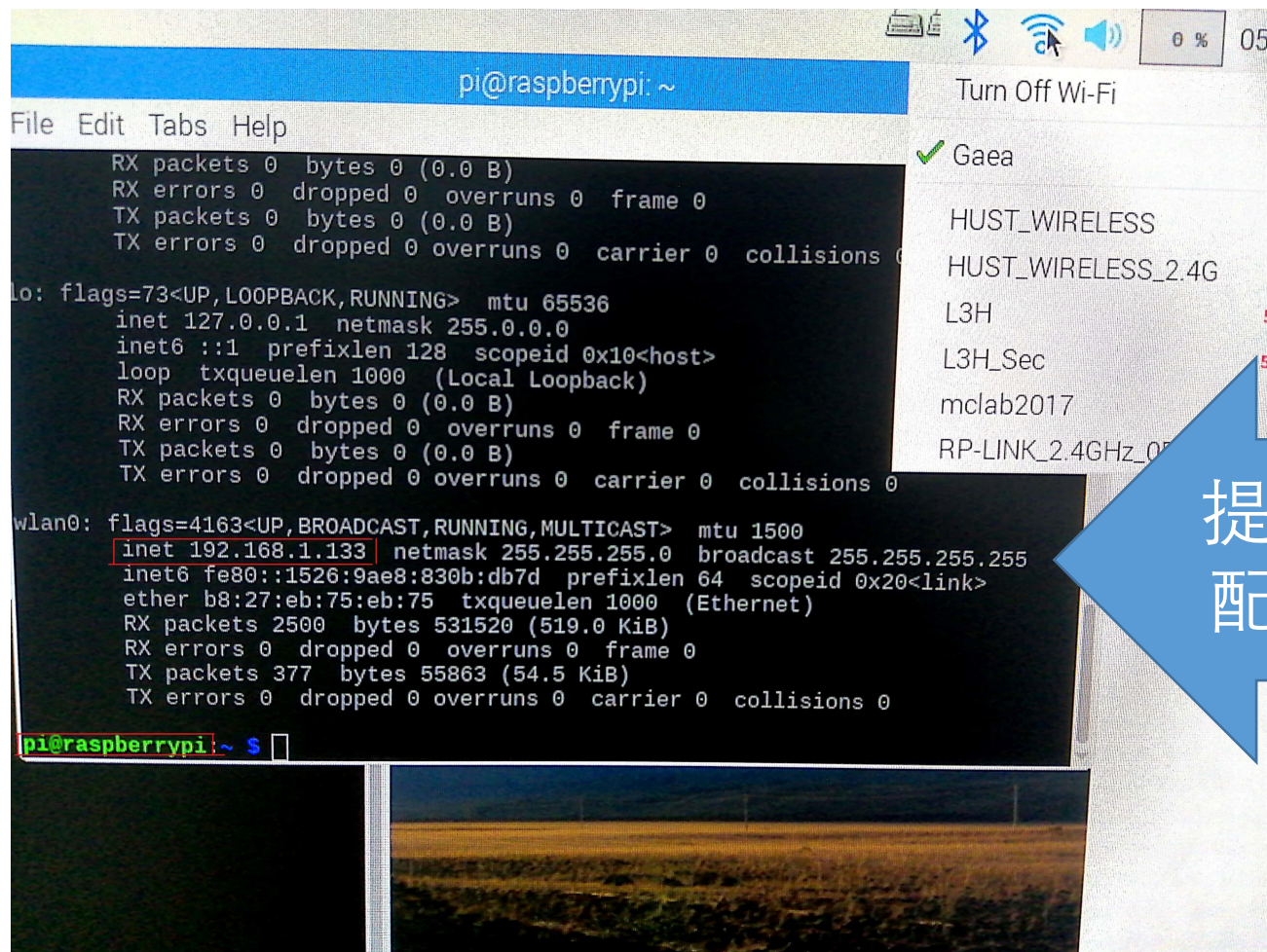
1.旋转

2.亮灯

3.显示器亮屏

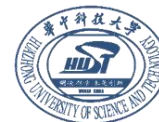


路由器通电，在显示器上为Burger小车连接网络。



连接结果

提示：可通过路由器设置对小车分配静态ip，以免之后修改配置文件



编辑-虚拟网络编辑器（左）。虚拟机-设置（右）。

虚拟网络编辑器

名称	类型	外部连接	主机连接	DHCP	子网地址
VMnet0	桥接模式	Intel(R) Wireless-AC 9560 16...	-	-	-
VMnet1	仅主机...	-	已连接	已启用	192.168.137.0
VMnet8	NAT 模式	NAT 模式	已连接	已启用	192.168.36.0

选择桥接模式

虚拟机设置

硬件 选项

设备 摘要

- 内存 2 GB
- 处理器 1
- 硬盘 (SCSI) 20 GB
- CD/DVD (SATA) 正在使用文件 E:\资料\计算机\...
- 网络适配器 桥接模式 (自动)
- USB 控制器 存在
- 声卡 自动检测
- 打印机 存在
- 显示器 自动检测

设备状态

- ☒ 已连接(C)
- ☒ 启动时连接(O)

网络连接

- ☒ 桥接模式(B): 直接连接物理网络
 - ☒ 复制物理网络连接状态(P)
- ☐ NAT 模式(N): 用于共享主机的 IP 地
- ☐ 仅主机模式(H): 与主机共享的专用网
- ☐ 自定义(U): 特定虚拟网络

VMnet0

LAN 区段(L):

LAN 区段(S)...

VMnet 信息

- ☒ 桥接模式(将虚拟机直接连接到外部网络)(B)
已桥接至(G): Intel(R) Wireless-AC 9560 160MHz #2
- ☐ NAT 模式(与虚拟机共享主机的 IP 地址)(N) NAT 设置(S)...
- ☐ 仅主机模式(在专用网络内连接虚拟机)(H)
- ☐ 将主机虚拟适配器连接到此网络(V)
主机虚拟适配器名称: VMware 网络适配器 VMnet0
- ☐ 使用本地 DHCP 服务将 IP 地址分配给虚拟机(D) DHCP 设置(P)...

子网 IP (I): . . . 子网掩码(M): . . .

选择自己的网卡

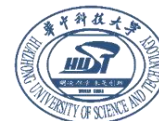
ssh 用户名@IP, 然后输入密码 (默认密码是turtlebot) 。

```
pi@raspberrypi: ~
File Edit Tabs Help
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 0 bytes 0 (0.0 B)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 0 bytes 0 (0.0 B)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
wlan0: flags=4163<UP,BROADCAST>
inet 192.168.1.133
inet6 fe80::1526:9ae8:
ether b8:27:eb:75:eb:75 txqueuelen 1000 (Ethernet)
RX packets 2500 bytes 531520 (519.0 KiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 377 bytes 55863 (54.5 KiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0
pi@raspberrypi:~$
```

```
shen@shen-virtual-machine:~/桌面$ ssh pi@192.168.1.133
pi@192.168.1.133's password:
Linux raspberrypi 4.19.36-v7+ #1213 SMP Thu Apr 25 15:08:02 B
ST 2019 armv7l

The programs included with the Debian GNU/Linux system are fr
ee software;
the exact distribution terms for each program are described i
n the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the ex
tent
permitted by applicable law.
Last login: Sat Mar 20 05:33:09 2021 from 192.168.1.106
pi@raspberrypi:~$ ls
2021-02-10-053453_640x480_scrot.png  Documents  Pictures
wget-log.1
2021-03-20-054429_1824x984_scrot.png  Downloads  Public
wget-log.2
2021-03-20-054442_1824x984_scrot.png  MagPi      Template
s
catkin_ws                             Music      Videos
Desktop                               opencr_update  wget-log
```



gedit ~/.bashrc，在末尾添加或修改为以下形式。【修改后重启终端】

【主机】：

```
export ROS_MASTER_URI = http://【主机 IP】:11311
```

```
export ROS_HOSTNAME = 【主机 IP】
```

```
export TURTLEBOT3_MODEL=burger
```

是为了在后续可视化操作中
调用正确的三维模型

【Burger】：

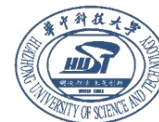
```
export ROS_MASTER_URI = http://【主机 IP】:11311
```

```
export ROS_HOSTNAME = 【Burger的IP】
```

Tips: **source ~/.bashrc**

这之后，显示器就基本没用了，可以关闭了。

提示：也可以
改成自动
ifconfig并获
取IP



在主机终端和主机ssh连接小车的终端上分别运行**roscore**，查看网络配置是否正确。

[如果不正确（如右图示），

在直接终端运行：

`export ROS_MASTER_URI=http://`

【主机 IP】:11311

`export ROS_HOSTNAME = 【Burger`

的IP】

再次**roscore**检查Burger地址设置是否正确。

或对`~/.bashrc`文件进行修改（注意

`source ~/.bashrc`）]

```
SUMMARY
=====

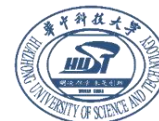
PARAMETERS
* /roscore: kinetic
* /rosversion: 1.12.14

NODES

WARNING: ROS_MASTER_URI [http://192.168.1.106:11311] host is not set to this machine
auto-starting new master
process[master]: started with pid [10311]
ROS_MASTER_URI=http://192.168.1.133:11311

setting /run_id to 48804b06-8a1e-11eb-8000-000000000000
process[roscout-1]: started with pid [10312]
started core service [/roscout]
```

本应该是
192.168.1.105



【小车终端】：运行 `roslaunch turtlebot3_bringup turtlebot3_robot.launch`

```
turtlebot3_core (roserial_python/serial_node.py)
turtlebot3_diagnostics (turtlebot3_bringup/turtlebot3
_diagnostics)
turtlebot3_lds (hls_lfcd_lds_driver/hlds_laser_publis
her)
```

```
ROS_MASTER_URI=http://192.168.1.105:11311
```

```
process[turtlebot3_core-1]: started with pid [1207]
process[turtlebot3_lds-2]: started with pid [1208]
process[turtlebot3_diagnostics-3]: started with pid [1209]
]
```

```
[INFO] [1616306224.308335]: ROS Serial Python Node
[INFO] [1616306224.349722]: Connecting to /dev/ttyACM0 at
115200 baud
```

```
[INFO] [1616306226.501496]: Note: publish buffer size is
1024 bytes
```

```
[INFO] [1616306226.504414]: Setup publisher on sensor_sta
te [turtlebot3_msgs/SensorState]
```

```
[INFO] [1616306226.528655]: Setup publisher on firmware_v
ersion [turtlebot3_msgs/VersionInfo]
```

```
[INFO] [1616306226.986045]: Setup publisher on imu [senso
r_msgs/Imu]
```

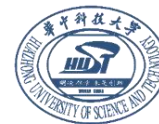
```
[INFO] [1616306227.012226]: Setup publisher on cmd_vel_rc
100 [geometry_msgs/Twist]
```

```
[INFO] [1616306227.190055]: Setup publisher on odom [nav_
msgs/Odometry]
```

```
[INFO] [1616306227.211039]: Setup publisher on joint_stat
es [sensor_msgs/JointState]
```

正常结果，正常IP

若出现异常，和rescore异常同样处理，也就是重新配置网络。



【主机终端】：新建两个终端，分别运行下列指令。

roslaunch turtlebot3_bringup turtlebot3_remote.launch

roslaunch turtlebot3_bringup turtlebot3_remote.launch
roslaunch rviz rviz -d `rospack find turtlebot3\_description`/rviz/model.rviz

```
roscore http://192.168.1.... x /home/shen/catkin_ws/s... x shen@shen-virtual-mach... x
shen@shen-virtual-machine:~$ roslaunch turtlebot3_bringup turtlebot3_remote.launch
... logging to /home/shen/.ros/log/57212548-8a20-11eb-a245-d9f6945a159b/roslaunch-shen-virtual-machine-22205.log
Checking log directory for disk usage. This may take a while.
Press Ctrl-C to interrupt
Done checking log file disk usage. Usage is <1GB.

xacro: in-order processing became default in ROS Melodic. You can drop the option.
started roslaunch server http://192.168.1.105:35939/

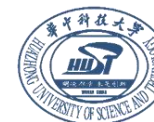
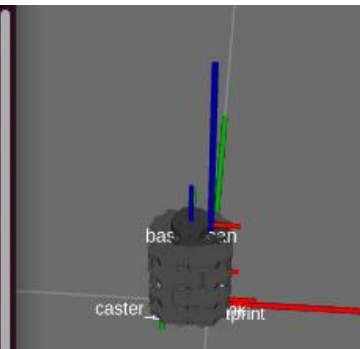
SUMMARY
=====

PARAMETERS
* /robot_description: <?xml version="1...
* /robot_state_publisher/publish_frequency: 50.0
* /robot_state_publisher/tf_prefix:
* /rostdistro: noetic
* /rosversion: 1.15.9
```

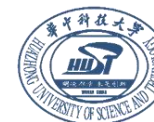
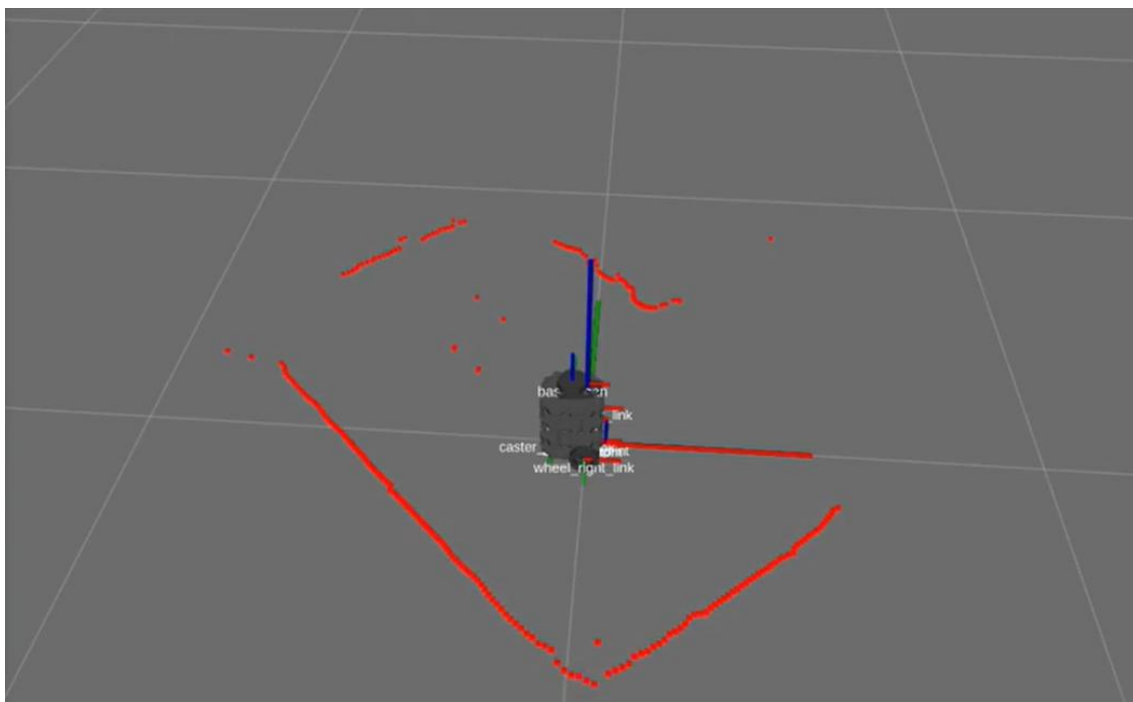
roslaunch

```
shen@shen-virtual-machine:~$ roslaunch rviz rviz -d `rospack find turtlebot3_description`/rviz/model.rviz
[ INFO] [1616315725.887973690]: rviz version 1.14.5
[ INFO] [1616315725.888722337]: compiled against Qt version 5.12.8
[ INFO] [1616315725.888884760]: compiled against OGRE version 1.9.0 (Ghadamon)
[ INFO] [1616315726.035188286]: Forcing OpenGL version 0.
[ INFO] [1616315728.514116505]: Stereo is NOT SUPPORTED
```

roslaunch



在**rviz**里我们可以看到一个虚拟仿真的小车。在程序窗口观察到调用的3D模型，窗口内出现的红点即为激光传感器的探测信息，描述了周围环境的光学反射点，通过对应的观察可以识别出椅子墙壁等物件。



【主机终端】：再新建一个终端，运行下列指令，开启远程键盘控制。

roslaunch turtlebot3_teleop turtlebot3_teleop_key.launch

```
ROS_MASTER_URI=http://192.168.1.105:11311
```

正常结果，正常IP

```
process[turtlebot3_teleop_keyboard-1]: started with p  
id [22390]
```

```
Control Your TurtleBot3!
```

```
-----  
Moving around:
```

```
      w      d  
a      s        
      x
```

可以用wasdx来控制真实小车以及rviz内的虚拟仿真小车

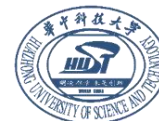
```
w/x : increase/decrease linear velocity (Burger : ~ 0  
.22, Waffle and Waffle Pi : ~ 0.26)
```

```
a/d : increase/decrease angular velocity (Burger : ~  
2.84, Waffle and Waffle Pi : ~ 1.82)
```

```
space key, s : force stop
```

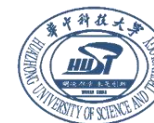
```
CTRL-C to quit
```

若真实小车控制成功，说明实验成功



The background features a large, light blue downward-pointing triangle. Overlaid on this are several geometric elements: a thin blue upward-pointing triangle, a thick blue downward-pointing triangle, and a blue arrow pointing towards the top right. The text "THANK YOU" is centered in a bold, blue, sans-serif font.

THANK YOU



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